

Prognosis and Clinical Course

Although *P. multocida* is a Gram-negative organism, most strains are susceptible to penicillin, which is the drug of choice if only *Pasteurella* is cultured. If one suspects that the bite reached the periosteum, parenteral penicillin should be given until the local lesion is well healed. *P. multocida* is also susceptible to doxycycline, later-generation cephalosporins, and trimethoprim-sulfamethoxazole.

Prevention

(Images removed)

RAT-BITE FEVER

AT A GLANCE

- Rat-bite fever refers to two clinically similar zoonoses caused by *Streptobacillus moniliformis* and *Spirillum minus*.
- Infection is usually acquired through a rat bite or ingestion of rat-contaminated food or drink.
- The disease is characterized by the classic triad of fever, polyarthralgias, and rash.

Etiology and Epidemiology

Rat-bite fever encompasses two zoonotic infections transmitted through contact with rats or their excreta. The more common form is caused by *Streptobacillus*

moniliformis, a pleomorphic Gram-negative rod found in the nasopharyngeal flora of most wild and laboratory rats, which serve as asymptomatic carriers. Carnivores that prey on wild rodents may also transmit the infection.

Infections with *S. moniliformis* occur via two routes:

1. Direct contact with infected animals, typically through a bite.
2. Ingestion of food or drink contaminated by rat urine, feces, or other secretions.

Both forms present with an acute flu-like illness featuring the clinical triad of fever, polyarthralgias or arthritis, and rash. An outbreak in Haverhill, Massachusetts, in 1926—caused by consumption of milk contaminated with rat excreta—led to the terms *Haverhill fever* or *erythema arthriticum epidemicum*.

The second causative agent is *Spirillum minus*, a Gram-negative spirochete. Spirillary rat-bite fever differs in geographic distribution, incubation period, and clinical features, with milder arthritis. It occurs almost exclusively in East Asia and is known in Japan as *sodoku*.

Clinical Findings

History

- **Streptobacillary disease:** Incubation period is typically 1 to 7 days, but may extend up to 3 weeks. The rat bite often has healed by the time symptoms begin. Illness starts abruptly with fever, chills, headache, stiff neck, and myalgias. A rash appears several days later.
- **Spirillary rat-bite fever:** Incubation period is usually longer than 14 days.

Cutaneous Lesions

- In **streptobacillary fever**, approximately 75% of patients develop a morbilliform rash 2 to 3 days after the onset of fever. Lesions may be macular, morbilliform, or petechial and are most prominent on the palms,

soles, and around joints, sometimes becoming generalized and resembling measles.

- The original bite site is typically healed and shows no signs of infection, inflammation, or regional adenopathy when the rash appears.
- About one week after resolution of the illness, desquamation of the palms and soles may occur, resembling the skin changes seen in Kawasaki disease.
- In **spirillary rat-bite fever**, the bite site is usually tender, red, indurated, or ulcerated when the rash develops.
- Cutaneous lesions in spirillary infection are larger and more prominent on the abdomen, resembling the rose spots of typhoid fever.